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```python

import numpy as np

# 4x4 matrix: rows represent students, columns represent subjects
student_scores = np.array([
    [85, 90, 78, 88],
    [92, 87, 85, 80],
    [76, 95, 82, 84],
    [88, 89, 91, 86]
])

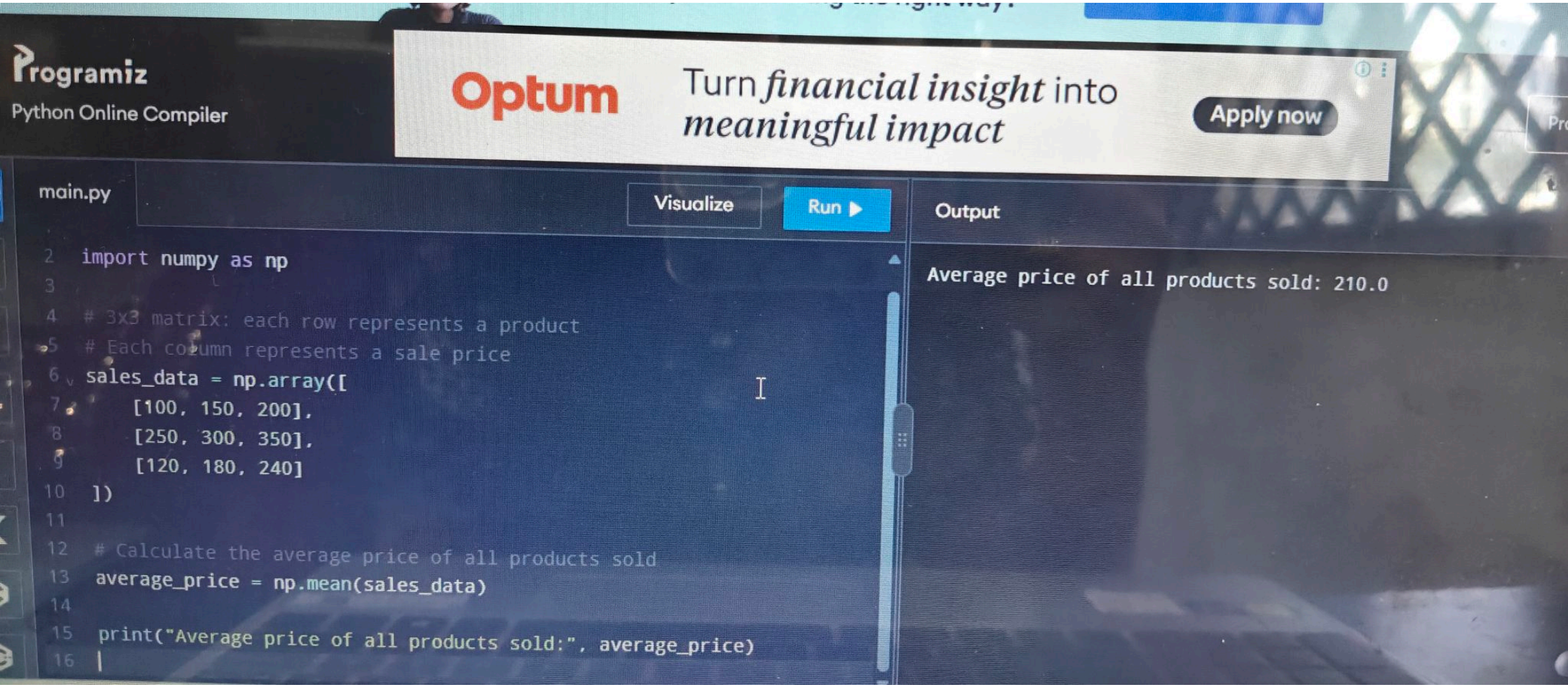
# Subject names in column order
subjects = ["Math", "Science", "English", "History"]

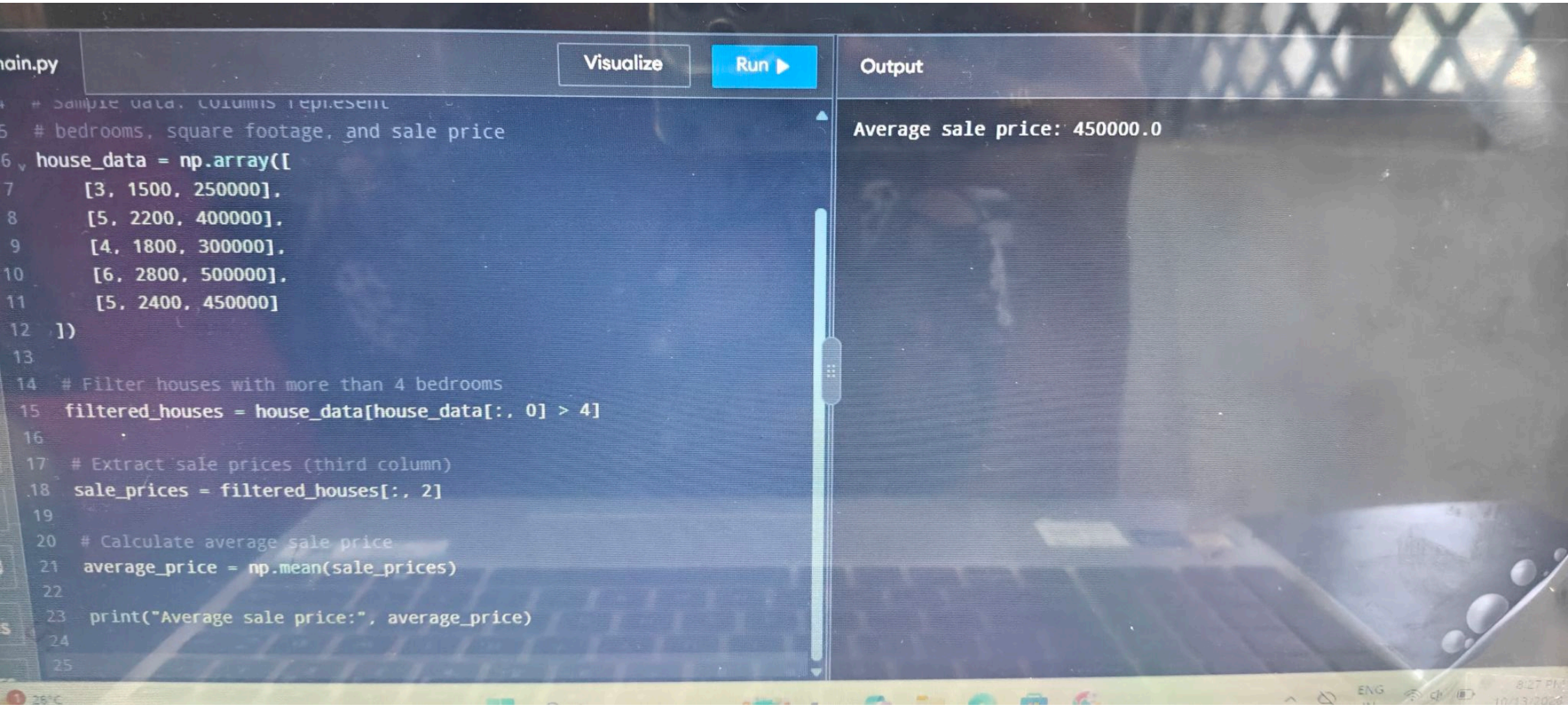
# Calculate average score for each subject (column-wise)
average_scores = np.mean(student_scores, axis=0)
```

Output

```
Math: 85.25
Science: 90.25
English: 84.00
History: 84.50

Subject with highest average: Science
Highest average score: 90.25
```


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import numpy as np
3x3 matrix: each row represents a product
Each column represents a sale price
sales_data = np.array([
[100, 150, 200],
[250, 300, 350],
[120, 180, 240]
])
Calculate the average price of all products sold
average_price = np.mean(sales_data)
print("Average price of all products sold:", average_price)
Average price of all products sold: 210.0


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Sample data: columns represent
bedrooms, square footage, and sale price
house_data = np.array([
[3, 1500, 250000],
[5, 2200, 400000],
[4, 1800, 300000],
[6, 2800, 500000],
[5, 2400, 450000]
])
Filter houses with more than 4 bedrooms
filtered_houses = house_data[house_data[:, 0] > 4]
Extract sale prices (third column)
sale_prices = filtered_houses[:, 2]
Calculate average sale price
average_price = np.mean(sale_prices)
print("Average sale price:", average_price)
Average sale price: 450000.0

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```python
import numpy as np

# Quarterly sales data (Q1, Q2, Q3, Q4)
sales_data = np.array([10000, 12000, 15000, 18000])

# Calculate total sales for the year
total_sales = np.sum(sales_data)

# Calculate percentage increase from Q1 to Q4
percentage_increase = (
    (sales_data[3] - sales_data[0]) / sales_data[0]
) * 100

# Display results
print("Total annual sales:", total_sales)
print("Percentage increase from Q1 to Q4:",
      percentage_increase, "%")

```

Output

```
Total annual sales: 55000
Percentage increase from Q1 to Q4: 80.0 %
```

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```
# Fuel efficiency of different car models (in miles per gallon)
fuel_efficiency = np.array([25, 30, 35, 40, 45])

# Calculate average fuel efficiency
average_efficiency = np.mean(fuel_efficiency)

# Compare two car models
model1 = fuel_efficiency[0]  # First car model: 25 MPG
model2 = fuel_efficiency[3]  # Fourth car model: 40 MPG

# Calculate percentage improvement
percentage_improvement = (
    (model2 - model1) / model1
) * 100

# Display results
print("Average fuel efficiency:", average_efficiency, "MPG")
print("Percentage improvement:", percentage_improvement, "%")
```

Output

```
Average fuel efficiency: 35.0 MPG
Percentage improvement: 60.0 %
```

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```
discount = subtotal * discount_rate / 100

# Calculate price after discount
discounted_total = subtotal - discount

# Calculate tax on discounted amount
tax = discounted_total * tax_rate / 100

# Calculate final total cost
total_cost = discounted_total + tax

# Display results
print("Subtotal:", subtotal)
print("Discount:", discount)
print("Amount after discount:", discounted_total)
print("Tax:", tax)
print("Final total cost:", total_cost)

```

Output

```
Subtotal: 850
Discount: 85.0
Amount after discount: 765.0
Tax: 38.25
Final total cost: 803.25
```